

III

Fourier Series

Periodic fn

$$f(x+T) = f(x) \quad T \text{ is the period.}$$

Even fn

$$f(x) = f(-x)$$

$$\text{eg: } y = |x|, x^2$$

Odd fn

$$f(-x) = -f(x)$$

$$y = x^3$$

$$\text{even} \times \text{even} = \text{even}$$

$$\text{even} \times \text{odd} = \text{odd}$$

$$\text{odd} \times \text{even} = \text{odd}$$

$$\text{odd} \times \text{odd} = \text{even}$$

$$\begin{aligned} \int_{-a}^a f(x) dx &= 0, \text{ if } f(x) \text{ is odd} \\ &= 2 \int_0^a f(x) dx, \text{ if } f(x) \text{ is even} \end{aligned}$$

$$\int u v dx = u_1 v_1 - u_1' v_2 + u_1'' v_3 - u_1''' v_4 + \dots$$

Let $f(x)$ be a fn defined in the interval $c \rightarrow c+2\pi$,

$f(x)$ can be represented as a Fourier series as follows;

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

a_0, a_n, b_n are Fourier coefficients given by

Euler's formula

$$a_0 = \frac{1}{\pi} \int_c^{c+2\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \sin nx dx$$

① Obtain ~~$f(x)$~~ the Fourier series representation for the function, $f(x) = \frac{(x-\pi)^2}{4}$, $0 < x < 2\pi$

Ans. $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$

where,

$$a_0 = \frac{1}{\pi} \int_c^{c+2\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \sin nx dx.$$

$$a. \quad f(x) = \frac{(\pi-x)^2}{4}, \quad 0 < x < 2\pi$$

$$a_0 = \frac{1}{4\pi} \int_0^{2\pi} (\pi-x)^2 dx = \frac{1}{4\pi} \left[-\frac{(\pi-x)^3}{3} \right]_0^{2\pi}$$

$$= \frac{1}{4\pi} \left[\frac{\pi^3}{3} + \frac{\pi^3}{3} \right] = \frac{\pi^2}{6} //$$

$$a_n = \frac{1}{4\pi} \int_0^{2\pi} (\pi-x)^2 \cos nx dx$$

$$= \frac{1}{4\pi} \left[(\pi-x)^2 \frac{\sin nx}{n} - (-2)(\pi-x) \left(\frac{\cos nx}{n^2} \right) + (-2)(-1) \left(\frac{\sin nx}{n^3} \right) \right]_0^{2\pi}$$

$$= \frac{1}{4\pi} \left[2(\pi-x) \frac{\cos nx}{n^2} + \cancel{2 \frac{\sin nx}{n^3}} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \left[(\pi-x) \frac{\cos nx}{n^2} \right]_0^{2\pi} = \frac{1}{2\pi} \left[\frac{\pi}{n^2} + \frac{\pi}{n^2} \right] = \frac{1}{n^2} //$$

$$b_n = \frac{1}{4\pi} \int_0^{2\pi} (\pi-x)^2 \sin nx dx$$

$$= \frac{1}{4\pi} \left[(\pi-x)^2 \frac{-\cos nx}{n} - (-2)(\pi-x) \left(\frac{\sin nx}{n^2} \right) + 2 \frac{\cos nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{4\pi} \left[(\pi-x)^2 - \frac{\cos nx}{n} + 2 \frac{\cos nx}{n^3} \right]_0^{2\pi}$$

$$\pi \frac{\cos nx}{n^2} - x \frac{\cos nx}{n^2}$$

$$= \frac{1}{4\pi} \left[\frac{-\pi^2}{n} + \frac{2}{n^3} - \left(\frac{-(\pi)^2}{n} + \frac{2}{n^3} \right) \right]$$

$$= \frac{\pi}{n^2} - \frac{2\pi}{n^2}$$

$$-\frac{\pi}{n^2} - \left(\frac{\pi}{n^2} \right)$$

$$= \frac{1}{4\pi} \left(\frac{-\pi^2}{n} + \frac{(\pi)^2}{n} \right) = \frac{1}{4\pi} \left(\frac{\pi^2}{n} - \frac{2\pi n + 1 - \pi^2}{n} \right)$$

$$= 0 //$$

$$f(x) = \frac{(\pi-x)^2}{4} = \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$$